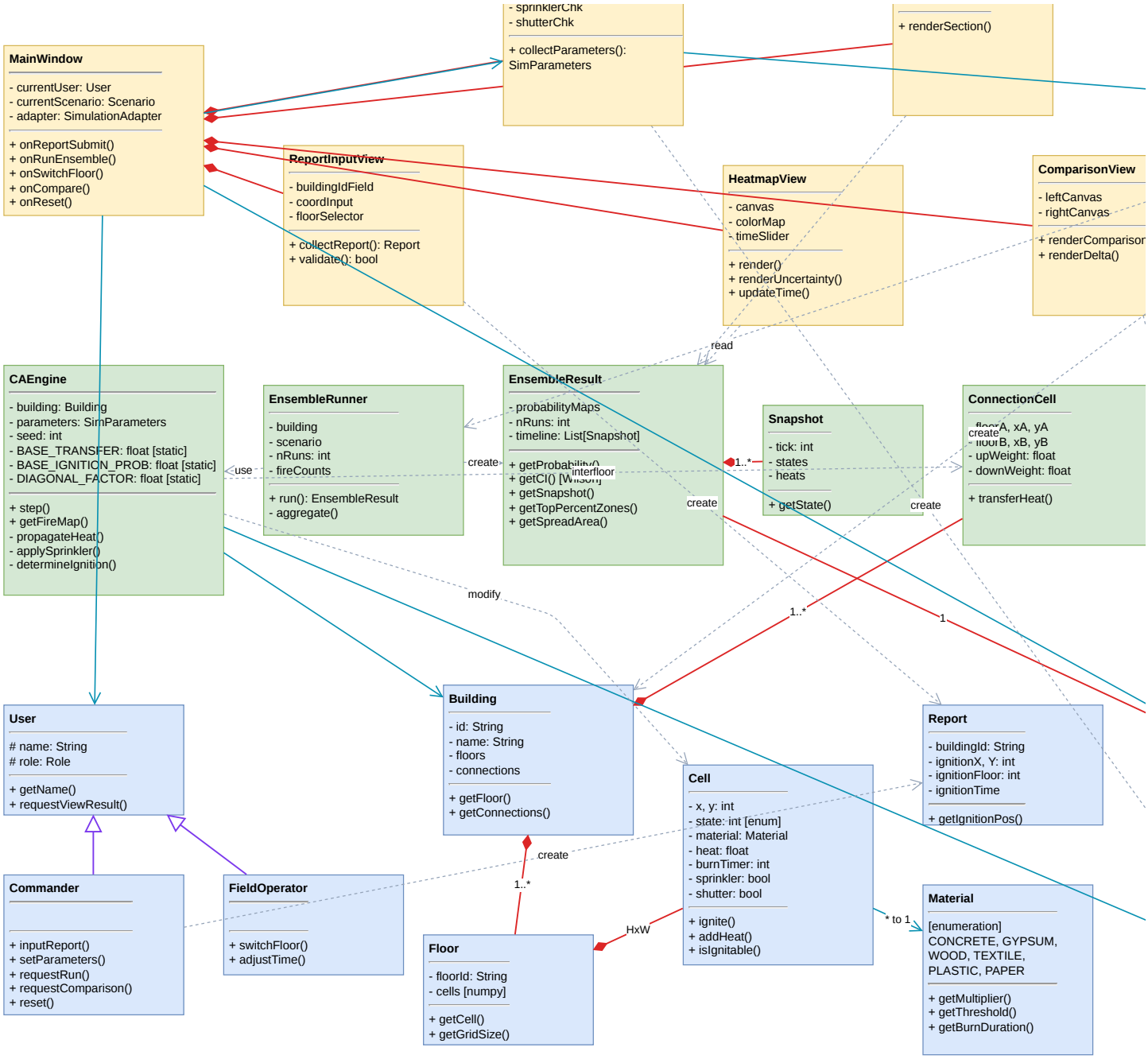
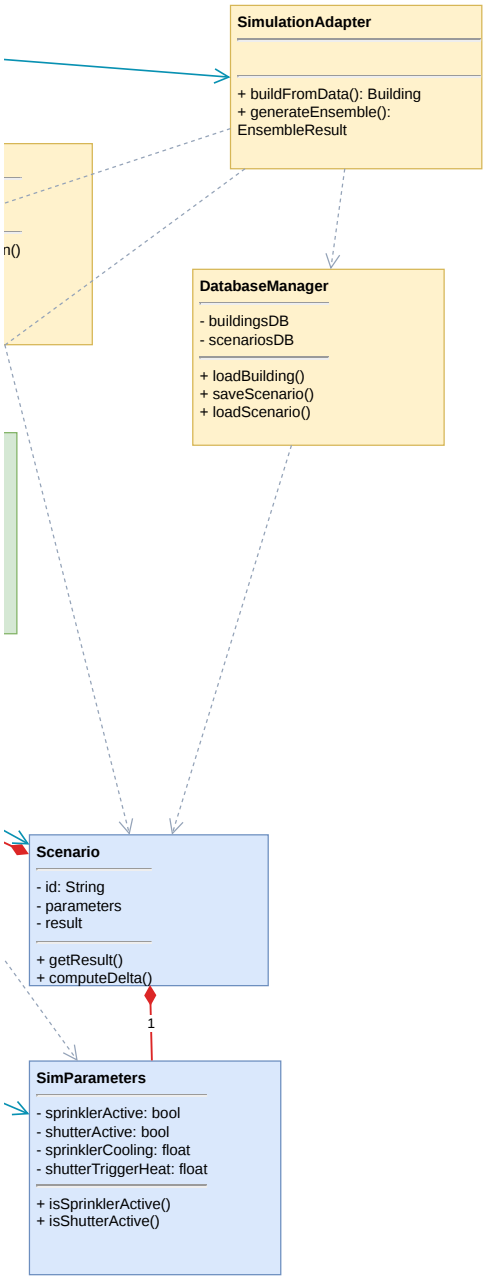
ParameterView

- sprinklerChk

SectionView

- canvas


MainWindow
- currentUser: User
- currentScenario: Scenario
- adapter: SimulationAdapter
+ onReportSubmit()
+ onRunEnsemble()
+ onSwitchFloor()
+ onCompare()
+ onReset()
- sprinklerChk
- shutterChk
+ collectParameters(): SimParameters
+ renderSection()
ReportInputView
- buildingIdField
- coordInput
- floorSelector
+ collectReport(): Report
+ validate(): bool
HeatmapView
- canvas
- colorMap
- timeSlider
+ render()
+ renderUncertainty()
+ updateTime()
ComparisonView
- leftCanvas
- rightCanvas
+ renderComparison
+ renderDelta()
read
CAEngine
- building: Building
- parameters: SimParameters
- seed: int
- BASE_TRANSFER: float [static]
- BASE_IGNITION_PROB: float [static]
- DIAGONAL_FACTOR: float [static]
+ step()
+ getFireMap()
- propagateHeat()
- applySprinkler()
- determineIgnition()
use
EnsembleRunner
- building
- scenario
- nRuns: int
- fireCounts
+ run(): EnsembleResult
- aggregate()
create
EnsembleResult
- probabilityMaps
- nRuns: int
- timeline: List[Snapshot]
+ getProbability()
+ getCI() [Wilson]
+ getSnapshot()
+ getTopPercentZones()
+ getSpreadArea()
interfloor
1..*
Snapshot
- tick: int
- states
- heats
+ getState()
create
ConnectionCell
- floorA, xA, yA
- floorB, xB, yB
- upWeight: float
- downWeight: float
+ transferHeat()
create
modify
1..*
1
User
name: String
role: Role
+ getName()
+ requestViewResult()
Building
- id: String
- name: String
- floors
- connections
+ getFloor()
+ getConnections()
Report
- buildingId: String
- ignitionX, Y: int
- ignitionFloor: int
- ignitionTime
+ getIgnitionPos()
Cell
- x, y: int
- state: int [enum]
- material: Material
- heat: float
- burnTimer: int
- sprinkler: bool
- shutter: bool
+ ignite()
+ addHeat()
+ isIgnitable()
create
1..*
HxW
* to 1
Commander
+ inputReport()
+ setParameters()
+ requestRun()
+ requestComparison()
+ reset()
FieldOperator
+ switchFloor()
+ adjustTime()
Floor
- floorId: String
- cells [numpy]
+ getCell()
+ getGridSize()
Material
[enumeration]
CONCRETE, GYPSUM, WOOD, TEXTILE, PLASTIC, PAPER
+ getMultiplier()
+ getThreshold()
+ getBurnDuration()


SimulationAdapter
+ buildFromData(): Building
+ generateEnsemble(): EnsembleResult
DatabaseManager
- buildingsDB
- scenariosDB
+ loadBuilding()
+ saveScenario()
+ loadScenario()
Scenario
- id: String
- parameters
- result
+ getResult()
+ computeDelta()
1
SimParameters
- sprinklerActive: bool
- shutterActive: bool
- sprinklerCooling: float
- shutterTriggerHeat: float
+ isSprinklerActive()
+ isShutterActive()